

Op-amp IC Tester

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Here is the circuit of an easy-to-use operational amplifier (op-amp) tester. This test gadget can be used to test the popular

general-purpose op-amp type 741 and other such op-amps that are pin-to-pin compatible to 741.

Op-amps are used in a large num-

ber of application circuits, so this test gadget can prove a great asset to hobbyists.

Circuit and working

In the circuit (Fig. 1), the op-amp under test can be wired either as an inverting amplifier with gain equal to $(R3/R4)$ or as a non-inverting amplifier with gain equal to:

$$1 + \left(\frac{R2}{R4}\right)$$

depending on the position of DPDT switch S2. The input to the amplifier in both the

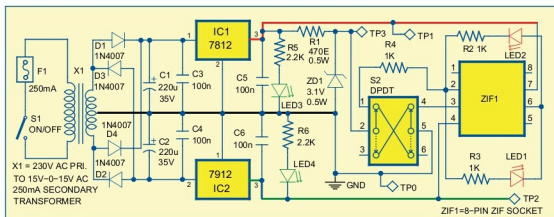


Fig. 1: Circuit of Op-amp IC tester

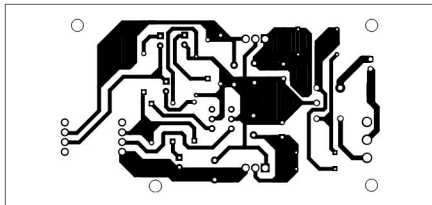


Fig. 2: An actual-size, single-side PCB for the op-amp tester

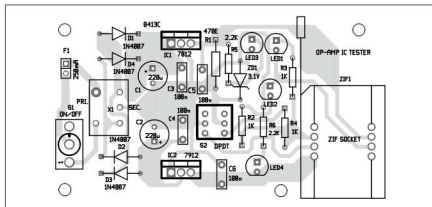


Fig. 3: Component layout for the PCB

Test Points

Test point	Details
TP0	0V, GND
TP1	12V
TP2	-12V
TP3	3.1V

cases is 3.1V DC. The circuit has its own regulated power supply to generate +12V and -12V for the op-amp from the mains power supply.

The power supply circuit consists of transformer X1 with a centre-tapped secondary winding followed by conventional full-wave rectifier using diodes D1 through D4. Rectifiers are followed by simple single-element capacitor filters. Diodes D1 and D4 constitute the full-wave rectifier for the positive supply, while the filtering action is provided by capacitor C1. Diodes D2 and D3, used with the same centre-tapped secondary winding, provide rectification for negative supply and capacitor C2 does the filtering.

IC1 is a +12V three-terminal regulator, while IC2 is a -12V three-